Chemical Reactions and Equations

Quick Revision

1. Chemical Reactions

- A chemical reaction is a change in which one or more substance(s) or reactant(s) react to form new substance(s) or product(s) with entirely different properties.
- The reacting species are known as **reactants** (the substances that undergo chemical change in the chemical reaction) and the new species formed as a result of the reaction are called **products**.
- A chemical reaction can be identified either by (i) change in state or colour (ii) evolution of gas (iii) change in temperature (iv) formation of a precipitate.

2. Chemical Equations

A chemical equation is the symbolic representation of a chemical reaction. Symbols and formulae of the reactants and products are used to write chemical equation. e.g.

$$\underset{\text{Methane}}{CH_4(g)} + \underset{\text{Oxygen}}{2O_2(g)} \xrightarrow{\Delta} \underset{\text{Carbon dioxide}}{CO_2(g)} + \underset{\text{Water}}{2H_2O(g)}$$

3. Balanced Chemical Equations

A balanced chemical equation is that in which the total number of atoms of each element are equal on both sides of the equation.

4. Types of Chemical Reactions

The chemical reactions are classified into different types depending upon the chemical changes taking place.

(i) **Combination Reaction** A reaction in which two or more reactants react together to form a single product, is called a combination reaction.

e.g. When calcium oxide (quick lime) is dissolved in water, it forms calcium hydroxide (slaked lime).

The reaction is highly exothermic, i.e. a lot of heat is produced during the reaction.

$$\underset{\text{Quicklime}}{CaO(s)} + \underset{\text{Water}}{H_2O(l)} \longrightarrow \underset{\text{Slaked lime}}{Ca(OH)_2(aq)}$$

(ii) **Decomposition Reaction** A reaction in which a single reactant breaks down to form two or more products, is known as decomposition reaction.

On the basis of the form of energy required for the reaction, these reactions are of three types

- **Thermal Decomposition** These reactions involve the use of energy in the form of heat for decomposition of the reactant. e.g.

$$\underset{\text{Calcium carbonate}}{CaCO_3(s)} \xrightarrow{\text{Heat}} \underset{\text{Calcium oxide}}{CaO(s)} + \underset{\text{Carbon dioxide}}{CO_2(g)}$$

- **Electrolysis** These reactions involve the use of electrical energy required for the decomposition of the reactant molecules. e.g.

$$\underset{\text{Water}}{2H_2O(l)} \xrightarrow{\text{Electric current}} \underset{\text{Hydrogen}}{2H_2(g)} + \underset{\text{Oxygen}}{O_2(g)}$$

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- **Photolysis or Photochemical Decomposition** These reactions involve the use of light energy for the purpose of decomposition. e.g.

e.g. $\underset{\text{Silver chloride}}{2AgCl(s)} \xrightarrow{\text{Sunlight}} \underset{\text{Silver}}{2Ag(s)} + \underset{\text{Chlorine}}{Cl_2(g)}$

(iii) **Displacement Reaction** This reaction is of two types :

- **Single Displacement Reaction** A reaction in which a more reactive element displaces a less reactive element from the solution of its compound, is called single displacement or displacement reaction.

e.g.

$$\underset{\text{Zinc metal (grey)}}{Zn(s)} + \underset{\text{Copper sulphate solution (blue)}}{CuSO_4(aq)} \longrightarrow \underset{\text{Zinc sulphate solution (colourless)}}{ZnSO_4(aq)} + \underset{\text{Copper metal (brown)}}{Cu(s)}$$

- **Double Displacement Reaction** The reaction in which two different ions in the reactant molecules are displaced by each other, is called double displacement reaction. e.g.

$$\underset{\text{Sodium sulphate}}{Na_2SO_4(aq)} + \underset{\text{Barium chloride}}{BaCl_2(aq)} \longrightarrow \underset{\text{Barium sulphate (white ppt.)}}{BaSO_4} + \underset{\text{Sodium chloride}}{2NaCl(aq)}$$

(iv) **Neutralisation Reaction** The reaction in which an acid reacts with a base to give salt and water is called neutralisation reaction. e.g.

$$\underset{\text{Sodium hydroxide}}{NaOH(aq)} + \underset{\text{Nitric acid}}{HNO_3(aq)} \longrightarrow \underset{\text{Sodium nitrate}}{NaNO_3(aq)} + \underset{\text{Water}}{H_2O(l)}$$

(v) **Oxidation and Reduction Reactions**

- **Oxidation** The process in which oxygen is added to a substance or the process in which hydrogen is removed from a substance is called oxidation.

$$\underset{\text{Sodium}}{4Na} + \underset{\text{Oxygen}}{O_2} \longrightarrow \underset{\text{Sodium oxide}}{2Na_2O},$$

$$\underset{\text{Ethanol}}{CH_3CH_2OH} \xrightarrow[300°C]{Cu} \underset{\text{Ethanal}}{CH_3CHO} + H_2$$

- **Reduction** The process in which hydrogen is added to a substance or the process in which oxygen is removed from a substance is called reduction.

$$\underset{\text{Sodium}}{2Na} + \underset{\text{Hydrogen}}{H_2} \longrightarrow \underset{\text{Sodium hydride}}{2NaH}$$

$$\underset{\text{Zinc oxide}}{ZnO} + \underset{\text{Carbon}}{C} \longrightarrow \underset{\text{Zinc}}{Zn} + \underset{\text{Carbon monoxide}}{CO}$$

- **Redox Reactions** Those reactions in which oxidation and reduction take place simultaneously are called redox reactions. In this reaction, the copper (II) oxide is losing oxygen and is being reduced.

Hydrogen is gaining oxygen and is being oxidised.

Reduction

$$\underset{\text{Copper oxide}}{CuO} + \underset{\text{Hydrogen}}{H_2} \longrightarrow \underset{\text{Copper}}{Cu} + \underset{\text{Water}}{H_2O}$$

Oxidation

- **Oxidising Agent** A substance which helps in the oxidation of another substance, is called oxidising agent. It is always to be remembered that oxidising agent itself gets reduced.
- **Reducing Agent** A substance that helps in the reduction of another substance, is called reducing agent. It is always to be remembered that reducing agent itself gets oxidised.

(vi) **Exothermic and Endothermic Reactions**

- **Exothermic reactions** The reactions which are accompanied by the evolution of heat, are called exothermic reactions. e.g.

$$\underset{\text{Methane}}{CH_4(g)} + \underset{\text{Oxygen}}{2O_2(g)} \longrightarrow \underset{\text{Carbon dioxide}}{CO_2(g)} + \underset{\text{Water}}{2H_2O(g)} + \text{Heat}$$

- **Endothermic reactions** The reactions which occur by the absorption of heat/energy are called endothermic reactions. e.g.

$$\underset{\text{Mercuric oxide}}{2HgO(s)} + \text{Heat} \longrightarrow \underset{\text{Mercury}}{2Hg(l)} + \underset{\text{Oxygen}}{O_2(g)}$$

5. Corrosion

The phenomenon due to which metals are slowly eaten away by the reaction of air, water and chemicals present in atmosphere, is called corrosion.

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- **Rusting of Iron** Corrosion of iron is called **rusting**. Iron forms hydrated iron oxide (rust) when kept open in moist air.

$$\underset{\text{Iron}}{4Fe} + \underbrace{3O_2 + xH_2O}_{\text{From air}} \longrightarrow \underset{\text{Hydrated iron oxide (rust)}}{2Fe_2O_3 \cdot xH_2O}$$

- **Formation of a Green Layer Over Copper** New copper (brown in colour) vessel forms a green layer of basic copper carbonate after some time due to the reaction of air and moisture.

$$\underset{\text{Copper}}{2Cu} + \underbrace{O_2 + CO_2 + H_2O}_{\text{From air}} \longrightarrow \underbrace{CuCO_3 \cdot Cu(OH)_2}_{\text{Basic copper carbonate (green coating)}}$$

- Corrosion causes damage to car bodies, bridges, iron railing, ships and all objects made of metals, specially those which are made of iron.
- Corrosion is a wasteful processs in most of cases. Every year, a lot of tones of various metals especially iron get wasted in the country. Hence, it is quite necessary to prevent corrosion.

6. Rancidity

- It is the process of slow oxidation of oils and fats present in the food materials resulting in the production of foul odour and taste in them.
- Anti-oxidants are added to foods containing fats and oils in order to prevent their oxidation.

Objective Questions

Multiple Choice Questions

01. Which one of the following process involve chemical reactions?

(a) Storing of oxygen gas under pressure in a gas cylinder
(b) Liquefaction of air
(c) Keeping petrol in a China dish in the open
(d) Heating copper wire in the presence of air at high temperature

02. The number of atoms of each element is conserved in any chemical reaction.

(a) True
(b) False
(c) Can't say
(d) Partially true/false

03. Complete the missing variables given as 'x' and 'y' in the following reaction:

$$Pb(NO_2)_2(aq) + 2KI(aq) \longrightarrow PbI_2(x) + KNO_3(y)$$

(a) aq and aq
(b) s and s
(d) aq and s
(d) s and aq

04. Identify x and y in the following reaction

$$Cu + xHNO_3 \longrightarrow Cu(NO_3)_2 + yNO_2 + 2H_2O$$

(a) 4 and 2 (b) 3 and 5
(c) 2 and 3 (d) 4 and 4

05. The following reaction is an example of

$$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(aq)$$

I. Displacement reaction
II. Combination reaction
III. Decomposition reaction
IV Exothermic reaction

(a) Only (III) (b) (I) and (III)
(c) (II) and (IV) (d) Only (I)

06. Ferrous sulphate crystals lose water when heated. How many molecules of water are lost?

(a) 5 (b) 7
(c) 2 (d) 4

07. Heating of limestone produces

(a) slaked lime (b) quick lime
(c) lime water (d) caustic soda

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08. Carbon dioxide gas burns with a pop sound when a burning candle is brought near it.
(a) True
(b) False
(c) Can't say
(d) Partially true/false

09. The formation of quick lime and carbon dioxide from calcium carbonate is an example of
(a) photochemical decomposition reaction
(b) thermal decomposition reaction
(c) oxidation reaction
(d) displacement reaction

10. Electrolysis of water is a decomposition reaction. The mole ratio of hydrogen and oxygen gases liberated during electrolysis of water is
(a) 1:1 (b) 2:1
(c) 4:1 (d) 1:2

11. Which of the following can be decomposed by the action of sunlight?
(a) KBr
(b) AgBr
(c) MgO
(d) NaCl

12. What will happen, when silver chloride is placed in sunlight for sometime?
(a) White silver chloride changes into grey coloured compound
(b) Decomposition takes place
(c) Both (a) and (b) takes place
(d) Nothing will happen

13. The heating of lead nitrate produces brown fumes of 'X' gas. Identify 'X' from the following.
(a) Lead oxide
(b) Oxygen
(c) Nitrogen dioxide
(d) Dinitrogen oxide

14. The carbonate of metal lead is a white solid. It decomposes when heated to form carbon dioxide and a yellow solid oxide 'X'. What is X?
(a) Zinc oxide (b) Lead oxide
(c) Silver oxide (d) Magnesium oxide

15. Which of the following is exothermic reaction?
(a) Dissolution of sodium hydroxide in water
(b) Evaporation of water
(c) Sublimation of silver chloride
(d) Dissolution of salt in water

16. Identify the endothermic process from the following:
(a) $H_2O(l) \longrightarrow H_2O(g)$
(b) $CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(aq)$
(c) Combustion of methane
(d) Addition of conc. HCl to water.

17. When zinc granules are reacted with dilute sulphuric acid, then which of the following gas is evolved?
(a) Oxygen gas
(b) Sulphur dioxide gas
(c) Hydrogen gas
(d) Hydrogen sulphide gas

18. Which of the following gives reddish-brown precipitate on mixing?
(a) $FeCl_3 + NH_4OH$ (b) $NaCl + H_2O$
(c) $AgNO_3 + NH_4OH$ (d) $CuSO_4 + H_2S$

19. The following reaction is an example of

$$Fe_2O_3 + 2Al \xrightarrow{\Delta} 2Fe + Al_2O_3$$

(a) thermal decomposition reaction
(b) displacement reaction
(c) double displacement reaction
(d) neutralisation reaction

20. Which of the following is incorrect statement?
(a) Mg is more reactive than H
(b) Fe is less reactive than Zn
(c) Cu is more reactive than Al
(d) Ag is less reactive than Cu

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21. What happens when copper rod is dipped in iron sulphate solution?

I. Copper displaces iron.
II. Blue colour of copper sulphate solution is obtained.
III. Reaction is exothermic.
IV. No reaction takes place.

(a) (I) and (II) (b) (I), (II) and (III)
(c) Only (I) (d) Only (IV)

22. What happens, when chlorine water is added to KI solution?

(a) $I_2(aq)$ is formed
(b) IO_3^- ions of white coloured are produced
(c) KOH is formed
(d) HCl(*g*) is evolved.

23. Match chemical compound given in the Column I with its colour given in Column II and select the correct answer using the options given below:

	Column I (Chemical compounds)		Column II (Colour of their compound)
A.	Barium sulphate	1.	Colourless
B.	Ferrous sulphate	2.	Blue
C.	Copper sulphate	3.	Green (light)
D.	Zinc sulphate	4.	White

Codes

	A	B	C	D
(a)	1	3	4	2
(b)	4	2	1	3
(c)	2	4	3	2
(d)	4	3	2	1

24. Complete the following reaction:

$$NaCl(aq) + AgNO_3(aq) \longrightarrow \ldots\ldots + \ldots\ldots$$

(a) Na, $Ag(NO_3)Cl$ (b) $NaNO_2$, AgOCl
(c) $NaNO_3$, $AgCl_2$ (d) $NaNO_3$, AgCl

25. The reaction in which two compounds exchange their ions to form two new compounds is

(a) precipitation reaction
(b) double displacement reaction
(c) decomposition reaction
(d) neutralisation reaction

26. In the double displacement reaction between aqueous potassium iodide and aqueous lead nitrate, a yellow precipitate of lead iodide is formed. While performing the activity if lead nitrate is not available, which of the following can be used in place of lead nitrate? *(NCERT Exemplar)*

(a) Lead sulphate (insoluble)
(b) Lead acetate
(c) Ammonium nitrate
(d) Potassium sulphate

27. In a double displacement reaction such as the reaction between sodium sulphate solution and barium chloride solution:

I. Exchange of atoms takes place.
II. Exchange of ions takes place.
III. A precipitate is produced.
IV. An insoluble salt is produced.

The correct option is *(CBSE 2020)*

(a) (II) and (IV) (b) (I) and (III)
(c) Only (II) (d) (III) and (IV)

28. The process of respiration is ..... .

(a) oxidation and exothermic
(b) reduction and endothermic
(c) oxidation and endothermic
(d) reduction and exothermic

29. When magnesium ribbon is burnt, which of the following statements is observed?

(a) Magnesium burns with white flame and changes into white powder
(b) Magnesium burns with white flame and changes into black powder
(c) Magnesium burns with yellow flame and changes into yellow powder
(d) Magnesium burns with yellow flame and changes into white powder

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30. Fire flies glow at night due to ..... of luciferin (protein).

(a) reduction (b) oxidation
(c) combustion (d) displacement

31. The addition of phosphorus and oxygen gives a product. This reaction is an example of ..... reaction.

(a) decomposition (b) precipitation
(c) oxidation (d) reduction

32. The following reaction is an example of which type of reaction:

$$2KClO_3(s) \xrightarrow{\Delta} 2KCl(s) + 3O_2(q)$$

(a) Reduction reaction
(b) Oxidation reaction
(c) Displacement reaction
(d) Exothermic reaction

33. Which of the following statements about the given reaction are correct?

$$2Fe(s) + 4H_2O(l) \longrightarrow Fe_3O_4(s) + 4H_2(g)$$

I. Iron metal is getting oxidised.
II. Water is getting reduced.
III. Water is acting as reducing agent.
IV. Water is acting as oxidising agent.

(a) (I), (II) and (III) *(NCERT Exemplar)*
(b) (III) and (IV)
(c) (I), (II) and (IV)
(d) (II) and (IV)

34. Identify the oxidising agent in the following:

$$MnO_2 + 4HCl \longrightarrow MnCl_2 + Cl_2 + 2H_2O$$

(a) MnO_2 (b) $MnCl_2$
(c) HCl (d) H_2O

35. In the equation,

$$Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + Cl_2 + 4H_2O,$$

the substance, Pb_3O_4 acts as

(a) an oxidising agent
(b) an acid
(c) a reducing agent
(d) a dehydrating agent

36. A dilute ferrous sulphate solution was gradually added to the beaker containing acidified permanganate solution. The light purple colour of the solution fades and finally disappears.

Which of the following is the correct explanation for the observation?

(a) $KMnO_4$ is an oxidising agent, it oxidises $FeSO_4$
(b) $FeSO_4$ acts as an oxidising agent and oxidises $KMnO_4$
(c) The colour disappears due to dilution, no reaction is involved
(d) $KMnO_4$ is an unstable compound and decomposes in the presence of $FeSO_4$ to a colourless compound

37. Identify the reducing agent in the following reaction:

$$H_2O + F_2 \longrightarrow HF + HOF$$

(a) H_2O (b) F_2
(c) HF (d) HOF

38. The following reaction is a type of ..... reaction.

$$PbO + H_2 \longrightarrow Pb + H_2O$$

(a) oxidation (b) reduction
(c) redox (d) decomposition

39. Match chemical reactions given in the Column I with the type of chemical reactions given in the Column II and select the correct answer using the options given below:

	Column I (Chemical reactions)		Column II (Types of chemical reactions)
A.	Formation of NH_3 from N_2 and H_2	1.	Decomposition
B.	Calcination of zinc carbonate	2.	Double displacement
C.	Rusting of iron	3.	Combination
D.	Reaction of aqueous $BaCl_2$ solution with dil. H_2SO_4	4.	Redox

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Codes

	A	B	C	D
(a)	3	1	4	2
(b)	3	4	2	1
(c)	1	4	3	2
(d)	4	3	2	1

40. Gold is a metal which does not corrode when exposed to the atmosphere.
(a) True
(b) False
(c) Can't say
(d) Partially true/false

41. The iron rod is covered with reddish-brown layer which damages the surface of rod. Which of the following method can be used to prevent its damage?
(a) By covered it with layer of base
(b) By covered it with thin layer of zinc
(c) By covered it with layer of dilute acid
(d) By covered its with layer of copper

42. Chemically, rust is .....
(a) hydrated ferrous oxide
(b) only ferric oxide
(c) hydrated ferric oxide
(d) None of the above

43. Food items made up of oils and fats are flushed with nitrogen gas
I. to protect them from being rancid.
II. to protect their taste.
III. to maintain the weight of food.
IV. to enhance their flavour.

The correct statements are
(a) (I), (II) and (III)
(b) (II) and (IV)
(c) (I) and (II)
(d) All of the above

44. ..... are used to prevent rancidity of food.
(a) Acids (b) Anti-oxidants
(c) Galvanisation items (d) Anti-reductants

45. Which of the following gases can be used for storage of fresh sample of an oil for a long time?
(a) Carbon dioxide or oxygen
(b) Nitrogen or oxygen
(c) Carbon dioxide or helium
(d) Helium or nitrogen

46. Match the chemical reactions given in the Column I with the gas evolved given in the Column II and select the correct answer using the options given below:

	Column I (Reactions)		**Column II** (Gas evolved)
A.	Zinc granules are treated with dilute H_2SO_4.	1.	O_2
B.	Calcium carbonate on reacting with HCl.	2.	CO_2
C.	Lead nitrate is heated strongly in a hard glass test tube.	3.	H_2
D.	Decomposition of $KClO_3$	4.	NO_2

Codes

	A	B	C	D
(a)	3	4	2	1
(b)	3	2	4	3
(c)	1	3	4	2
(d)	3	2	4	1

Assertion-Reasoning MCQs

Direction (Q.Nos. 47-54) For question numbers two statements are given–one labeled **Assertion** (A) and the other labelled **Reason** (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

(a) Both A and R are true and R is correct explanation of the A.
(b) Both A and R are true but R is not the correct explanation of the A.
(c) A is true but R is false.
(d) A is false but R is true.

47. Assertion The following chemical equation,

$$2C_6H_6 + \frac{7}{2}O_2 \longrightarrow 4CO_2 + 3H_2O$$

is a balanced chemical equation.

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Reason In a balanced chemical equation, the total number of atoms of each element is equal on both side of the equation.

48. Assertion $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

Reason Photochemical decomposition occurs in the presence of sunlight.

49. Assertion Quicklime reacts vigorously with water releasing a large amount of heat.

Reason The above chemical reaction is an exothermic reaction.

50. Assertion Photosynthesis is considered as an endothermic reaction.

Reason Energy gets released in the process of photosynthesis.

51. Assertion $Fe_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Fe$

The above chemical equation is an example of displacement reaction.

Reason Aluminium being more reactive than iron, displaces Fe from its oxide.

52. Assertion Reaction between silver nitrate and potassium chloride is a precipitation reaction.

Reason Precipitation reaction produces insoluble salt.

53. Assertion In the following chemical equation,

$CuO(s) + Zn(s) \longrightarrow ZnO(s) + Cu(s)$

Zinc is getting oxidised and copper oxide is getting reduced.

Reason The process in which oxygen is added to a substance is called oxidation whereas the process in which oxygen is removed from a substance is called reduction.

54. Assertion Corrosion involves the damage or deterioration of metal components.

Reason The reduction reaction occurs between metal and atmospheric gases.

Case Based MCQs

55. Read the following and answer questions from (i) to (v).

Chemical reactions involve the breaking and making of bonds between atoms to produce new substances. During a chemical reaction atoms of one element do not change into those of another element. Nor do, atoms disappear from the mixture or appear from elsewhere. There are certain types of reactions. Reactions in which a single product is formed from two or more reactants is known as a combination reactions.

Decomposition reactions are the reaction in which a compound breaks down into simpler compounds.

Displacement and double displacement reactions are one in which an atom or group of atom is replaced by another. A double displacement reaction usually occurs in solution and one of products, being insoluble, percipitate out (separate as a solid). Another of reaction is redox reactions in which simultaneous oxidation and reduction takes place.

(i) Which of the following reactions involved the combination of two element?

(a) $CaO + CO_2 \longrightarrow CaCO_2$

(b) $4Na + O_2 \longrightarrow 2Na_2O$

(c) $SO_2 + \frac{1}{2}O_2 \longrightarrow SO_3$

(d) $NH_3 + HCl \longrightarrow NH_4Cl$

(ii) Consider the reaction

$Fe_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Fe$

The above reaction is an example of

(a) combination reaction

(b) double displacement reaction

(c) decomposition reaction

(d) simple displacement reaction

(iii) The equation,

$Mg(s) + CuO(s) \longrightarrow MgO(s) + Cu(s)$

represents

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